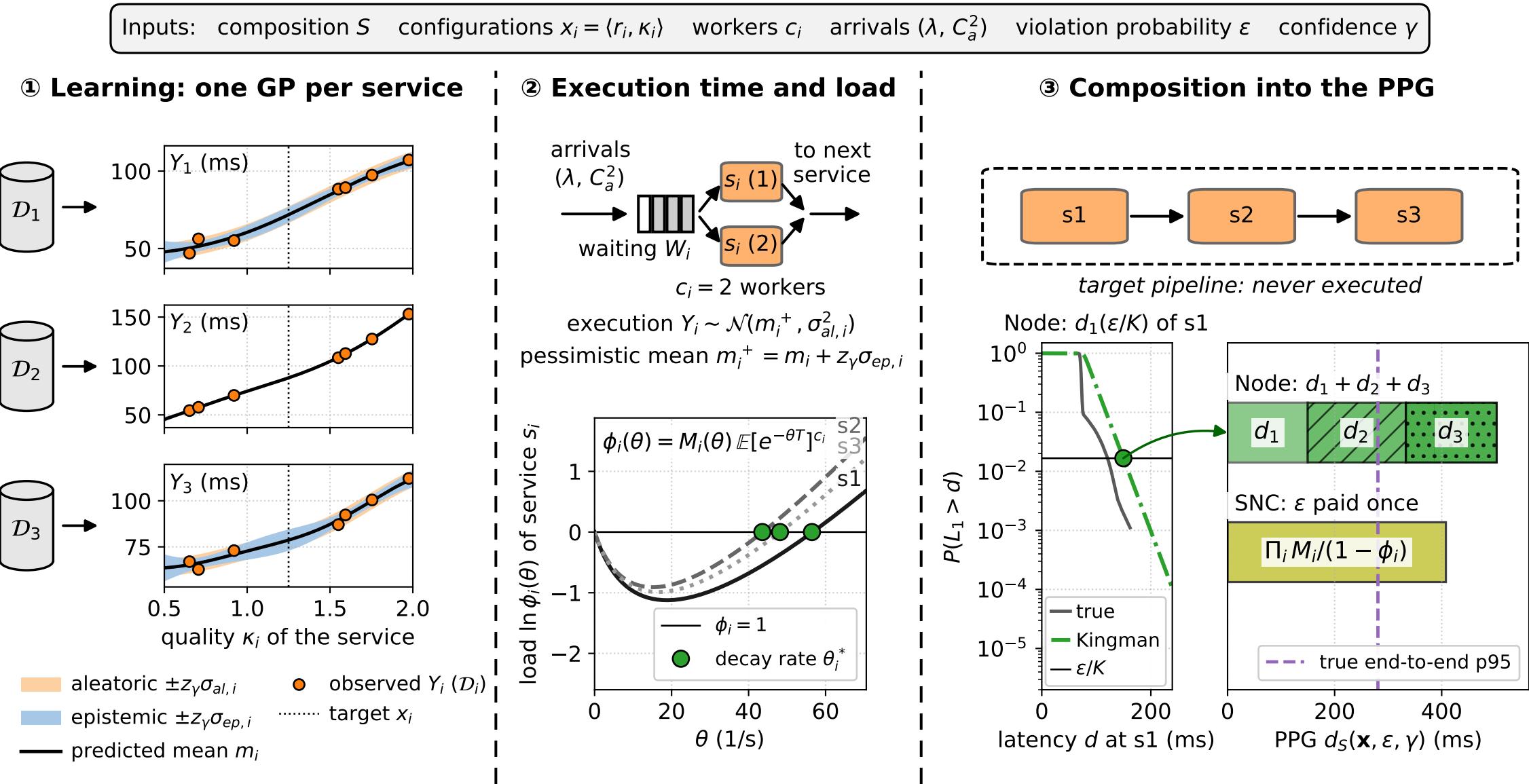

Inputs: composition S configurations $x_i = \langle r_i, \kappa_i \rangle$ workers c_i arrivals (λ, C_a^2) violation probability ε confidence γ
① Learning: one GP per service
$\mathcal{D}_1$
$\mathcal{D}_2$
$\mathcal{D}_3$
Y_1 (ms)
Y_2 (ms)
Y_3 (ms)
quality κ_i of the service
aleatoric $\pm z_\gamma \sigma_{al,i}$
epistemic $\pm z_\gamma \sigma_{ep,i}$
predicted mean m_i
observed Y_i ($\mathcal{D}_i$)
target x_i
② Execution time and load
arrivals (λ, C_a^2)
waiting W_i
s_i (1)
s_i (2)
to next service
$c_i = 2$ workers
execution $Y_i \sim \mathcal{N}(m_i^+, \sigma_{al,i}^2)$
pessimistic mean $m_i^+ = m_i + z_\gamma \sigma_{ep,i}$
$\phi_i(\theta) = M_i(\theta)\,\mathbb{E}[e^{-\theta T}]^{c_i}$
load $\ln \phi_i(\theta)$ of service s_i
θ (1/s)
$\phi_i = 1$
decay rate θ_i^*
③ Composition into the PPG
s1
s2
s3
target pipeline: never executed
Node: $d_1(\varepsilon/K)$ of s1
$P(L_1 > d)$
latency d at s1 (ms)
true
Kingman
ε/K
Node: $d_1 + d_2 + d_3$
d_1
d_2
d_3
SNC: ε paid once
$\Pi_i M_i/(1-\phi_i)$
true end-to-end p95
PPG $d_S(\mathbf{x}, \varepsilon, \gamma)$ (ms)